

04/24/2024

Certificate of Tests

PAGE: 6

Purchase Order: 26627161

Sales Order: 1001000647

Heat Number

Vendor Name

Mfg. Date

22173121

Big River Steel - Osceola AR US

01/11/2023

Item Number

Description

11000102

5L- X52 4-1/2X.337 (15.00#)BR,DRL,PEB

CHEMICAL ANALYSIS

	C	Mn	P	S	Ni	Cr	Cu	Mo	Si	Ti	V	CE
Ladle	0.0500	0.5400	0.0120	0.0030	0.0500	0.0600	0.1300	0.0120	0.1400	0.0000	0.0020	0.0935
	Al:	0.0270	Sn:	0.0048	N:	0.0068	B:	0.0001	Ca:	0.0023	Nb:	0.0120
Chem	0.1040	0.5500	0.0150	0.0040	0.0500	0.0610	0.1280	0.0110	0.1540	0.0010	0.0030	0.1484
	Al:	0.0310	Sn:	0.0050	N:	0.0160	B:	0.0001	Ca:	0.0029	Nb:	0.0130
Chem	0.0980	0.5500	0.0150	0.0030	0.0500	0.0610	0.1280	0.0120	0.1570	0.0010	0.0030	0.1426
	Al:	0.0320	Sn:	0.0050	N:	0.0140	B:	0.0001	Ca:	0.0023	Nb:	0.0140

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 69.90 TS (ksi): 74.10

EL (%): 28.00 HRB: 84.00 Y/T(%): 0.94

CHARPY TEST RESULTS BODY

ORIENTATION: TRANSVERSE
Spec. Size (in) : .394"X.197"X2.165"
Test Temp.(of) 0
Location 3T90

Specimen(S)	Energy (ft/lbs.)	%Shear
1	79.00	100.00
2	81.00	100.00
3	80.00	100.00
Average	80.00	100.00

Acceptance Criteria Min. FT/lbs: Min: Ft/Lbs 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Big River Steel - Osceola AR US
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept